The First Revolution

I began graduate school at Oxford University in the fall of 1984, and within a few months the corridors were abuzz with talk of a revolution in physics. As the Internet was yet to be widely used, rumor was a dominant channel for the rapid spread of information, and every day brought word of new breakthroughs. Researchers far and wide commented that the atmosphere was charged in a way unseen since the early days of quantum mechanics, and there was serious talk that the end of theoretical physics was within reach.

String theory was new to almost everyone, so in those early days its details were not common knowledge. We were particularly fortunate at Oxford: Michael Green had recently visited to lecture on string theory, so many of us became familiar with the theory's basic ideas and essential claims. And impressive claims they were. In a nutshell, here is what the theory said:

Take any piece of matter—a block of ice, a chunk of rock, a slab of iron—and imagine cutting it in half, then cutting one of the pieces in half again, and on and on; imagine continually cutting the material into ever smaller pieces. Some 2,500 years ago, the ancient Greeks had posed the problem of determining the finest, uncuttable, indivisible ingredient that would be the end product of such a procedure. In our age we have learned that sooner or later you come to atoms, but atoms are not the answer to the Greeks' question, because they can be cut into finer constituents. Atoms can be split. We have learned that they consist of electrons that swarm around a central nucleus that is composed of yet finer particles—protons and neutrons. And in the late 1960s, experiments at the Stanford Linear Accelerator revealed that even neutrons and protons themselves are made up of more fundamental constituents: each proton and each neutron consists of three particles known as quarks, as mentioned in Chapter 9 and illustrated in Figure 12.3a.

Conventional theory, supported by state-of-the-art experiments, envisions electrons and quarks as dots with no spatial extent whatsoever; in this view, therefore, they mark the end of the line—the last of nature's matryoshka dolls to be found in the microscopic makeup of matter.